

JITHESWAR

AI / ML Software Engineer

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EDUCATION

B.Tech, Computer Science and Engineering

Expected 2028

KL University, Hyderabad

EXPERIENCE

Research Software Engineer Intern

DigiClinics | Feb 2026 - Aug 2026

- Built a plant/herb classification pipeline combining **TF-IDF retrieval** with **INT8 quantization** for efficient on-device inference, reaching a cross-validation score of **0.9864**.
- Pursuing the NVIDIA NCA-ADS (AI in Data Science) certification alongside internship work.

PROJECTS

Liquid Glass An Omarchy desktop theme whose palette is sampled from its own wallpaper; blur, squircles, and per-app alpha. 15 stars and 3 forks on GitHub.

SIH26038 Explainable AI for diabetic retinopathy screening, Smart India Hackathon 2026. Calibrated ICDR grading with Grad-CAM evidence cross-checked against classical lesion detection; sensitivity 0.982, specificity 0.917, sealed external test set unopened.

Agent Spend Policy Engine Autonomous agent spend controls enforced with real x402 payments on Algorand testnet. Policy, not trust.

Contextual Priority Platform Re-prioritises container CVEs by what the Kubernetes cluster is actually doing, using runtime reachability with EPSS and CISA KEV rather than static CVSS. Built solo as an academic team submission.

NIDS Zero-trust network intrusion detection with AI-based anomaly detection, mTLS, Keycloak SSO, and hash-chained audit logging.

Resume Screening and Ranking Two-stage candidate screening that explains every position and ranks on redacted resumes. It advises; it never decides.

Additional: Ayur-Ai / HerbRAG (offline Ayurvedic leaf classifier with TF-IDF herb retrieval, extending the DigiClinics work) · medical imaging classifiers for pneumonia, lung fibrosis and malaria (EfficientNet-B3, MobileNetV2, Grad-CAM) · ruralAI (rural health platform: LLM symptom checker, prescription OCR, offline-first mobile app) · FSAD-Project (Next.js, NestJS, Prisma, Redis) · API-Pulse · local-ai-companion (hybrid vector + BM25 memory on Ollama)

TECHNICAL SKILLS

Languages: Python, TypeScript, Java, MATLAB, Shell

ML / AI: PyTorch, CNNs (EfficientNet, MobileNetV2), TF-IDF & BM25 retrieval, hybrid dense retrieval, INT8 quantization, Grad-CAM explainability, model calibration, RAG pipelines, Ollama

Backend: FastAPI, Spring Boot, NestJS, Prisma, Redis

Frontend: Next.js, TypeScript

Infrastructure / Security: Docker, Kubernetes, AWS, mTLS, Keycloak SSO, JWT auth, zero-trust architecture, container vulnerability triage (EPSS, CISA KEV)

Tools: Git/GitHub, Linux (Arch, Hyprland), Google Colab, Jupyter